

Algebraic and Transcendental functions

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Last compilation: November 12, 2025

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November 13, 2025

Algebra skills

Use the properties of logarithms to condense the following logarithmic expression. To show the use of each property, write each step of the process.

$$\frac{1}{5} [\log_8 (y) + 2 \log_8 (y + 4) - \log_8 (y - 1) + \log_8 (\sqrt[3]{y})]$$

Combine the functions $f(x) = x^2 + 1$ and $g(x) = x - 4$, according to the indicated operation,

- $(f - g)(x)$
- $(f \cdot g)(-4)$

Solve the following equations,

(a) $4^{x+1} \ln(30) = 10$

(b) $e^{x^2+6} - e^x = 0$

Function analysis

Determine algebraically whether the two given functions are the inverse of each other or not. Justify your answer with $f(g(x))$ and $g(f(x))$,

$$f(x) = 1 + x^3, \quad g(x) = \sqrt[3]{1+x}.$$

Find all the asymptotes, x and y interceptions of the following function, domain and range,

$$f(x) = \frac{x+6}{5+3x}$$

Modelling with math

Gym subscription. A monthly subscription for a gym cost is \$380 if the package is for 6 months or less. If you buy a subscription for more than 7 months up to 12 months, the monthly cost reduces to \$300, and if you buy a subscription for any period longer than a year, the monthly cost further reduces to \$250. Determine the piecewise function that corresponds to the situation and sketch its graph.

Algebra Toolkit

Algebra properties

$$a + b = b + a,$$

$$(a + b) + c = a + (b + c),$$

$$a(b + c) = ab + ac,$$

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}, \quad \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd},$$

$$\frac{-a}{b} = -\frac{a}{b} = \frac{a}{-b},$$

$$\frac{\frac{a}{b}}{\frac{c}{d}} = \frac{ad}{bc}$$

$$\sqrt{ab} = \sqrt{a}\sqrt{b}, \quad \sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}, \quad \sqrt[n]{a} = a^{1/n}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a},$$

$$a^2 - b^2 = (a - b)(a + b)$$

Laws of exponents

$$a^r \cdot a^s = a^{r+s}, \quad (a^r)^s = a^{rs}$$

$$\frac{a^r}{a^s} = a^{r-s}, \quad (ab)^r = a^r b^r$$

$$\left(\frac{a}{b}\right)^r = \frac{a^r}{b^r}, \quad b \neq 0$$

Laws of Logarithms

$$\log_a(AB) = \log_a(A) + \log_a(B)$$

$$\log_a\left(\frac{A}{B}\right) = \log_a(A) - \log_a(B)$$

$$\log_a(A^C) = C \log_a(A)$$

Inverse functions property

$$(f \circ f^{-1})(x) = x = (f^{-1} \circ f)(x)$$

$$\log_a(a^x) = x = a^{\log_a(x)}$$

Radioactive decay. A sample of a radioactive substance has an initial mass of 763.3 mg. This substance follows a continuous exponential decay model and has a half-life of 5 days.

- Which is the decay rate (as percentage of the tenth)?
- How much will be present in 22 days?
- How much time will it take to reach 60 mg

Conic sections

Identify the type of conic section and find the following,

- | | |
|-------------------|---------------------|
| • If is a circle, | • If is an ellipse |
| 1. Center | 1. Center |
| 2. Radius | 2. Focus (2) |
| | 3. Semi-axis (2) |
| • If is parabola | • If is a hyperbola |
| 1. Vertex | 1. Center |
| 2. Focus | 2. Focus (2) |
| 3. Directrix | 3. Asymptotes (2) |

1. $-16x^2 + 9y^2 - 128x + 18y - 391 = 0$

2. $\frac{x^2}{16} - \frac{y^2}{9} = 1$

Conic Sections

Standard form of a circle

$$(x-h)^2 + (y-k)^2 = r^2$$

Standard form of a parabola

$$(x-h) = 4a(y-k)^2, \text{ or } (y-k) = 4a(x-h)^2$$

Standard form of an ellipse

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

Standard form of a hyperbola

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$$